

Financial Simulation & Strategy Report

A Monte Carlo Analysis of the Trinity Strategy with S&P; 500 over 30 Years

Date of Generation: September 06, 2026

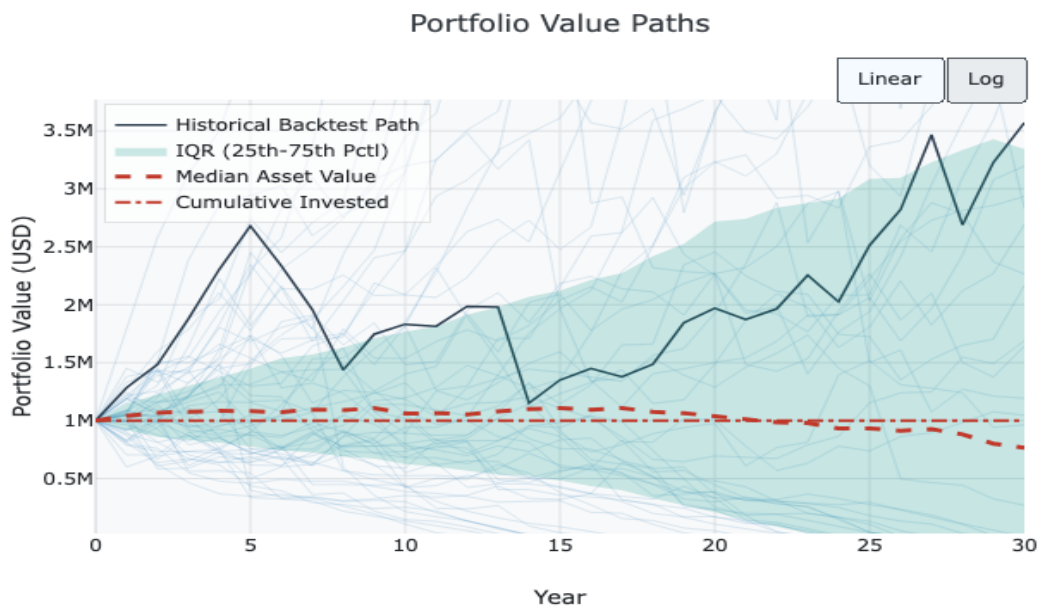


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1. Important Disclaimer

Important Disclaimer: Not Financial Advice

This document and its contents are for informational and educational purposes only and do not constitute financial, investment, legal, or tax advice. The simulations and analyses presented herein are based on historical data and mathematical models, which have inherent limitations.

No Guarantee of Future Performance: Past performance, whether actual or simulated, is not indicative of future results. The future performance of any asset or financial strategy is unknown and subject to a wide range of market, economic, and other risks. There is no guarantee that any of the outcomes projected in this report will be realized.

Simulation-Based Analysis: The results in this report are the output of a Monte Carlo simulation, which is a tool for modeling the probability of different outcomes. The simulation relies on assumptions about asset returns, volatility, and other economic factors that may not hold true in the future. The range of outcomes presented does not represent all possible scenarios.

Consult a Professional: You should not make any financial decisions based solely on the information contained in this report. Before making any investment or financial planning decisions, you should consult with a qualified and licensed financial advisor who can assess your individual circumstances and risk tolerance.

The author and publisher of this report disclaim all liability for any direct or indirect loss or damage arising from the use of, or reliance on, the information and analysis contained herein.

2. Executive Summary

The Scenario: This report analyzes a "Trinity" strategy using S&P; 500, starting with \$1,000,000 over 30 years.

Analysis was disabled.

Key Statistics	
Final Net Worth (Real Nominal)	\$422,932 \$766,084
Total Portfolio Value (Real Nominal)	\$2,045,655 \$2,388,807 (Final: \$766,084 + Withdrawn: \$1,622,723)
Return on Investment	138.9% (2.9% annualized)
Portfolio Success Rate	63.0%
Total Withdrawn	\$1,622,723
Years of Spending Left (Portfolio Longevity)	10.8 years remaining
Total Costs	\$433,228 (Tax: \$346,582 Fees: \$86,645)
■■ Cost Drag	15.4% of value (27% of withdrawals)
Chance of Ruin	37.0%
Worst 5% Outcome	\$0
Earliest Ruin Year	Year 4 (in worst 37% of cases)

Portfolio Health Dashboard



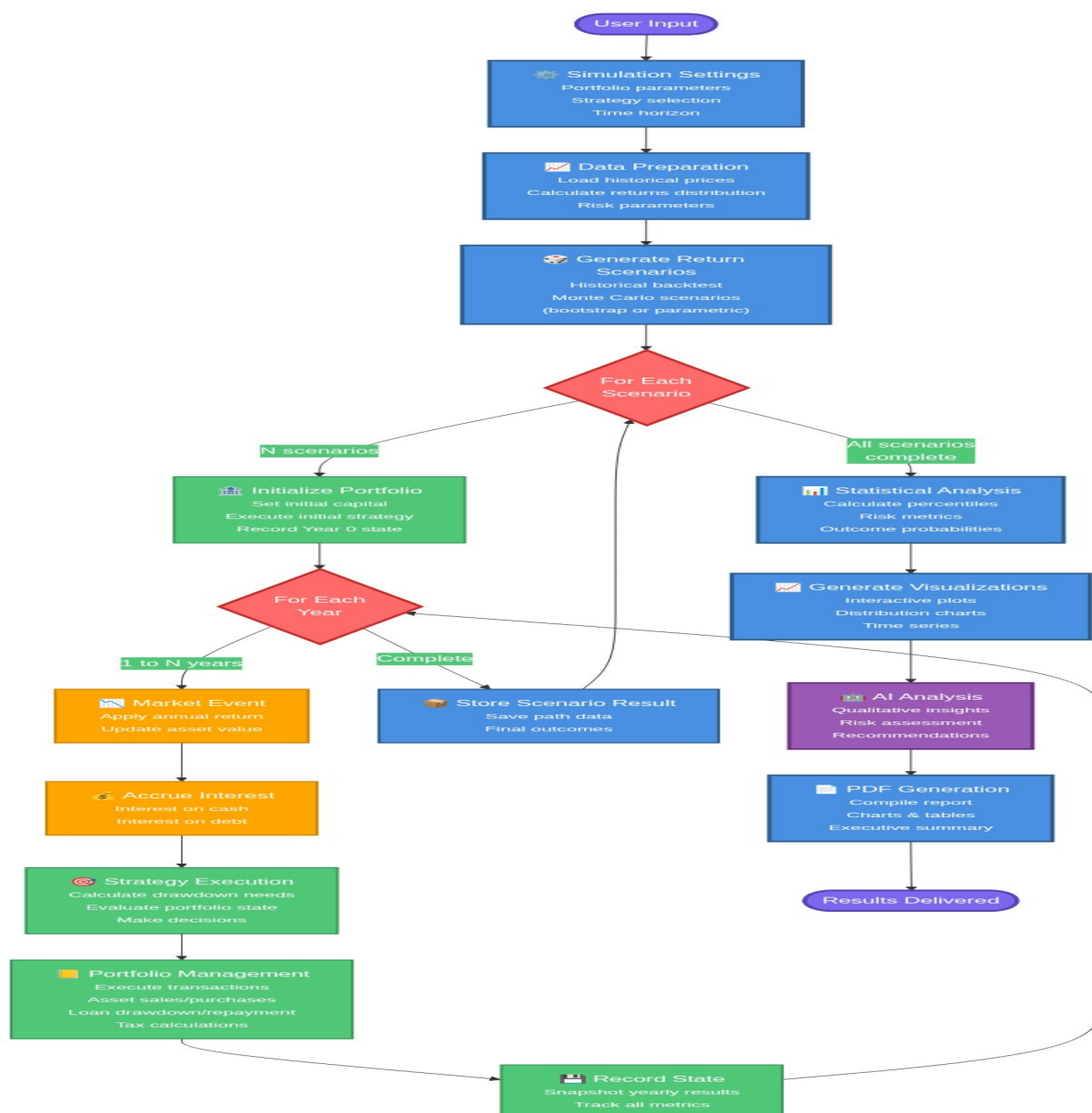
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3. Methodology Overview

This Monte Carlo simulation evaluates financial strategies for generating liquidity from a portfolio over multi-decade timeframes. By running thousands of scenarios with randomized market returns, we can quantify the range of potential outcomes and assess the risk-reward profile of different approaches such as **Asset Withdrawal** (Trinity Study) and **Loan Drawdown** (Buy, Borrow, Die).

Simulation Process Flow

The following flowchart illustrates the step-by-step process of the Monte Carlo simulation:



Stage 1: Asset Return Modeling

This stage defines the asset's return characteristics, which form the basis for the simulation. Two

models are available:

- **Bootstrap Model:** This model loads historical price data for an asset (e.g., BTC/USD or a stock index). It then computes the historical distribution of rolling annual returns. This method is designed to capture the asset's authentic historical volatility and return profile, including phenomena such as "fat tails" (extreme events) and sequence risk. The simulation then samples directly from these historical returns.
- **Parametric Model:** This model simulates returns based on user-defined parameters for expected annual return and volatility. Daily returns are drawn from a normal distribution, a standard assumption in financial modeling for assets like a global index fund. This allows for testing scenarios that may differ from historical performance.

Stage 2: Monte Carlo Simulation Engine

This is the core of the analysis, where thousands of parallel "possible futures" are simulated to assess the range of outcomes. The simulation operates using two nested loops:

Scenario Loop (Outer): For each of the N scenarios (e.g., 10,000 simulations):

- Initialize a fresh portfolio with starting capital
- Execute the initial strategy setup (Year 0)
- Enter the Year Loop

Year Loop (Inner): For each year in the time horizon, the following events occur sequentially:

1. Market Event: The asset's performance for the year is determined. In 'Bootstrap' mode, an annual return is randomly selected from the historical data. In 'Parametric' mode, 365 days of small, random price changes are simulated based on the specified parameters. The portfolio's asset value is updated accordingly.
2. Interest Accrual: Interest is calculated and applied to both cash holdings (positive interest) and outstanding debt (interest costs).
3. Strategy Execution: Each simulation uses a **Strategy** (the decision-making logic) to determine what actions to take. The strategy evaluates the current portfolio state, historical performance, and mandatory costs to decide on the optimal course of action.

Available Actions: Strategies can issue commands to perform various actions, including:

- **Sell Assets:** Liquidate a portion of the portfolio to generate cash (used by the Trinity/Withdrawal strategy)
- **Take Loan:** Borrow money against the portfolio value (used by the Buy, Borrow, Die strategy)
- **Repay Debt:** Pay down outstanding loans to reduce leverage
- **Add Contributions:** Inject new capital into the portfolio
- **Suspend Drawdowns:** Temporarily halt consumption withdrawals based on risk thresholds

Built-in Strategies: The simulation includes pre-built strategies like "Asset Withdrawal" (Trinity Study approach) and "Loan Drawdown" (Buy, Borrow, Die). However, custom strategies can be designed to combine any of these actions based on sophisticated logic, market conditions, or risk metrics.

4. Portfolio Management: A **Portfolio Manager** (the execution and bookkeeping engine) executes

the strategy's commands while maintaining accurate records of all transactions, costs, taxes, and portfolio state. This includes:

- Executing asset sales and purchases
- Processing loan drawdowns and repayments
- Calculating and applying transaction costs
- Computing capital gains taxes
- Tracking cost basis for tax purposes

5. Risk Management (LTV Thresholds): For loan-based strategies, a tiered Loan-to-Value (LTV) management system can be enabled to prevent insolvency. This includes a **Warning Tier** that proactively suspends new loan drawdowns if LTV exceeds a lower threshold (e.g., 50%), and an **Action Tier** that triggers emergency deleveraging (forced asset sales to repay debt) if LTV breaches a higher, more critical threshold (e.g., 70%). The portfolio manager executes these risk management actions automatically when triggered by the strategy.

6. State Recording: At the end of each year, the portfolio's complete state is recorded, including Asset Value, Debt, Net Worth, Cash Position, and all transaction details. This creates a comprehensive year-by-year history for each simulation path.

After completing all years in the Year Loop, the final scenario results are stored, and the process repeats for the next scenario.

Stage 3: Results Analysis & Insights

Following the completion of all simulations, this stage consolidates the results to provide meaningful insights:

- Statistical Aggregation: The data from all simulations is aggregated, and key statistics are calculated. To provide a robust view of the "typical" outcome and its variability, this analysis focuses on the **median** (50th percentile) and the **Interquartile Range (IQR)**, which represents the 25th to 75th percentile of outcomes. These metrics are less sensitive to extreme outliers than the mean and standard deviation, making them more suitable for analyzing the often-skewed distributions found in financial modeling.
- Visualization Generation: Interactive plots and charts are created to illustrate the range of outcomes, including time-series projections, distribution histograms, and comparative analyses.
- AI-Powered Analysis: An AI model reviews the quantitative results to provide qualitative insights, risk assessments, and strategic recommendations. This narrative analysis helps interpret the "why" behind the numbers and offers context for decision-making.
- PDF Report Compilation: All findings, visualizations, and analyses are compiled into a comprehensive PDF report for review and archiving.

4. Strategic Analysis: Asset Withdrawal vs. Loan Drawdown

Financial strategies in this simulation operate by executing various types of transactions to achieve their goals. Understanding the available transaction types is essential for interpreting how a strategy works and why it produces certain outcomes. This chapter first introduces the general toolkit of transactions available to any strategy, then describes the specific approach used in this simulation.

Available Transaction Types

Available Actions	
Asset Sales	Selling assets to generate cash for consumption, costs, or debt repayment. Triggers capital gains tax.
Asset Purchases	Buying assets with available cash to increase portfolio value and future growth potential.
Borrowing	Taking loans against portfolio value to fund expenses without selling assets. Incurs interest costs.
Debt Repayment	Paying down existing debt to reduce interest costs and improve the loan-to-value ratio.
Contributions	Adding new capital to the portfolio from external sources (e.g., salary, savings).
Cash Buffer Management	Holding cash reserves to fund withdrawals in down markets, reducing the need to sell assets at unfavorable prices.
Deleveraging	Strategy-triggered asset sales when LTV ratios exceed thresholds, used to reduce debt and restore healthy leverage ratios.

Strategy Used in This Simulation

Trinity Study Strategy (Asset Withdrawal)

This is a traditional decumulation strategy based on the famous Trinity Study. The strategy operates by systematically selling assets each year to fund living expenses.

Core Mechanics:

- Each year, a predetermined percentage of the initial portfolio value is withdrawn
- The withdrawal amount is adjusted annually for inflation
- All cash needs (consumption + costs) are funded by selling assets

- No debt is ever taken

Primary Goal: Ensure the portfolio is not depleted before the end of the time horizon. Success means the portfolio provides the desired cash flow for the entire period, even if the final value is less than the starting value.

Key Risk: Portfolio depletion, especially from "sequence of returns risk" where poor returns in early years can disproportionately harm the portfolio's longevity.

Transaction Pattern: This strategy uses only Asset Sales and Cost Payments. It never borrows, contributes additional capital, or purchases assets after the initial investment.

Strategy Decision Flow

The following diagram illustrates simplified annual decision-making logic of this strategy:



Note: This strategy has not yet been evaluated on the leaderboard. Evaluation data will appear here once the strategy is tested.

5. Simulation Settings

This section provides a detailed breakdown of all the input parameters used for this simulation run. It serves as a precise record of the scenario being tested, covering everything from the general simulation setup to the specific rules governing the chosen financial strategy and asset model.

General	
Initial Investment	1,000,000
Num Simulations	1,000
Num Years	30
Simulation Name	Trinity 4% On S&P; 500 (Browser E2E)

Strategy	
Strategy	Trinity Study (Asset Withdrawal)
Annual withdrawal rate as a percentage of the initial portfolio value.	4.00%

Asset Model	
Asset Model	S&P; 500 (^GSPC)
Asset Management Fee	0.25%
Return Threshold Rate	None

Economic Assumptions	
Cash Interest Rate	1.00%
Inflation Rate	2.00%
Loan Interest Rate	5.00%

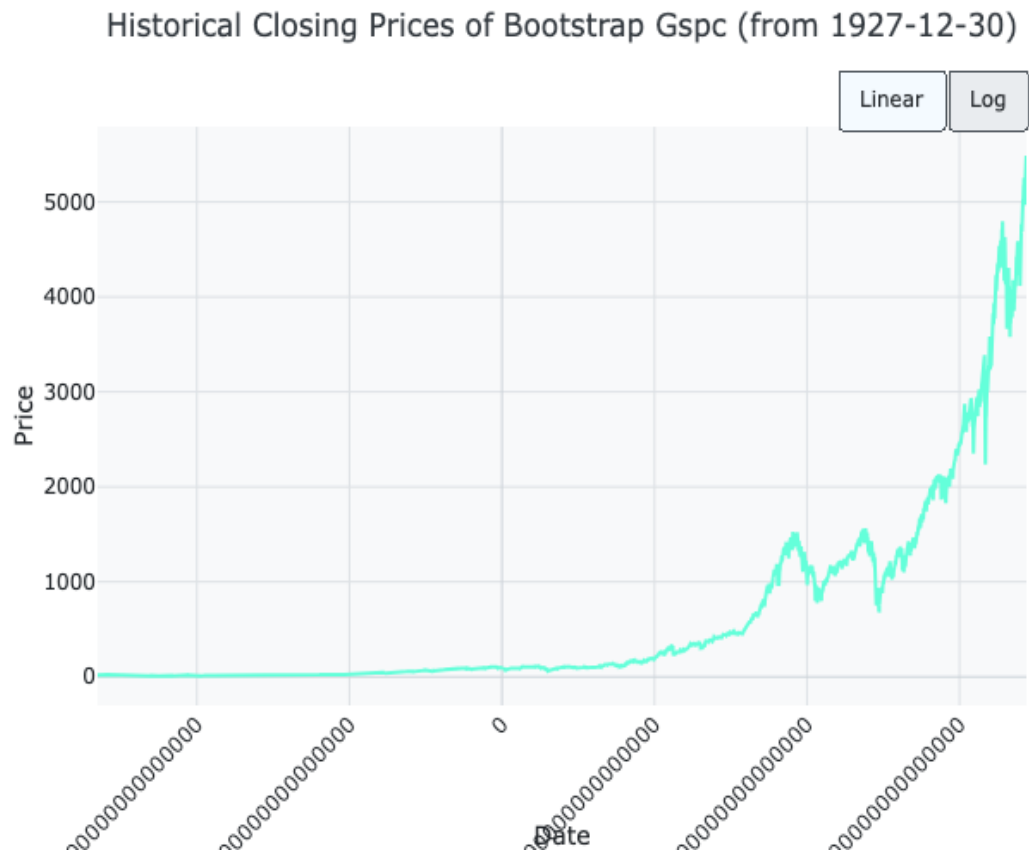
Tax	
Isk Tax Rate	1.00%
Capital Gains Tax Rate	30.00%
Tax Method	Isk

6. Input Data Analysis

Before delving into the complexities of multi-year financial simulations, it is paramount to thoroughly understand the foundational data upon which these simulations are built. This chapter is dedicated to dissecting the historical characteristics of the chosen asset. By examining its price history, return distributions, and historical drawdowns, we can better contextualize the simulation's outcomes. For detailed technical analysis including rolling returns distributions and autocorrelation charts, see the 'Input Data Analysis' appendix. > ****FX Risk Disclosure:**** This simulation uses the historical returns of the underlying asset directly. No dynamic currency conversion (e.g., USD/SEK exchange rate volatility) is modeled. The results assume that the asset's performance in its native currency translates directly to your portfolio currency, effectively assuming a constant exchange rate or a perfectly hedged position. Actual results may differ significantly due to currency fluctuations.

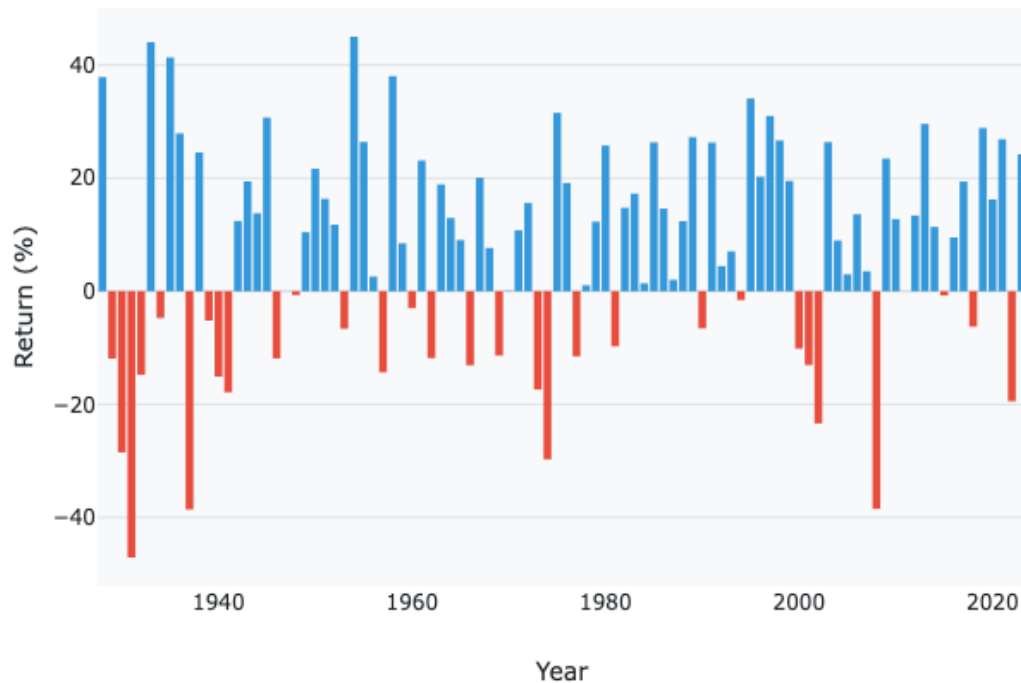
Asset Information

Asset Information	
Asset Name	S&P; 500
Ticker	^GSPC
Start Date	1927-12-30
End Date	2024-06-24
Total Trading Days	24,235
CAGR	6.12%
Annualized Volatility	18.93%
Sharpe Ratio	0.41
Sortino Ratio	0.58
Maximum Drawdown	-86.19%
Calmar Ratio	0.07
Skewness	-0.12
Kurtosis	17.26

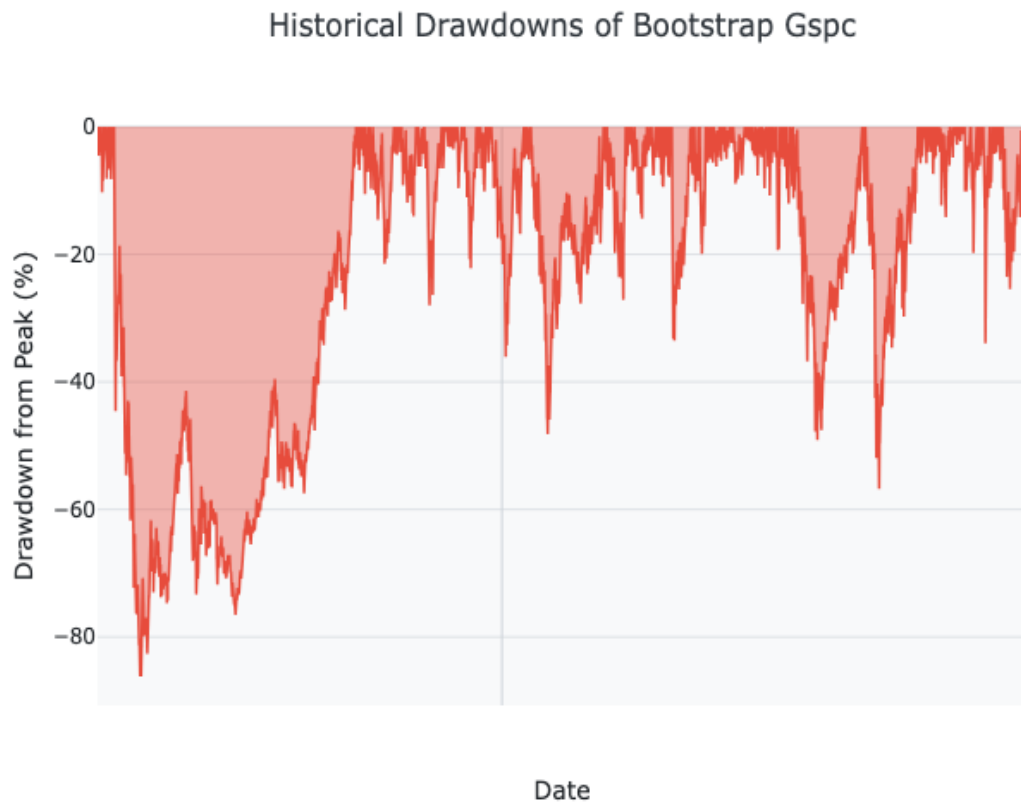


This chart presents the historical price trajectory of the asset over the selected period using a logarithmic y-axis. This scaling is particularly useful for assets with exponential growth, as it helps to visualize the relative magnitude of price changes over time. On a log scale, constant percentage growth appears as a straight line, offering a clearer perspective on the asset's compound growth rate and volatility. It provides a macroscopic view of the asset's long-term performance, illustrating its growth, periods of consolidation, and significant bull and bear markets.

Historical Calendar Year-over-Year Bootstrap Gspc Returns



This bar chart displays the asset's total return for each calendar year in the dataset. It offers a clear and immediate impression of the asset's year-over-year volatility and performance. The fluctuations between positive and negative returns highlight the 'sequence of returns risk'—the danger that a series of poor returns in the early stages of a withdrawal strategy can disproportionately harm a portfolio's longevity. This plot is essential for understanding the boom-and-bust cycles that can characterize investment returns.

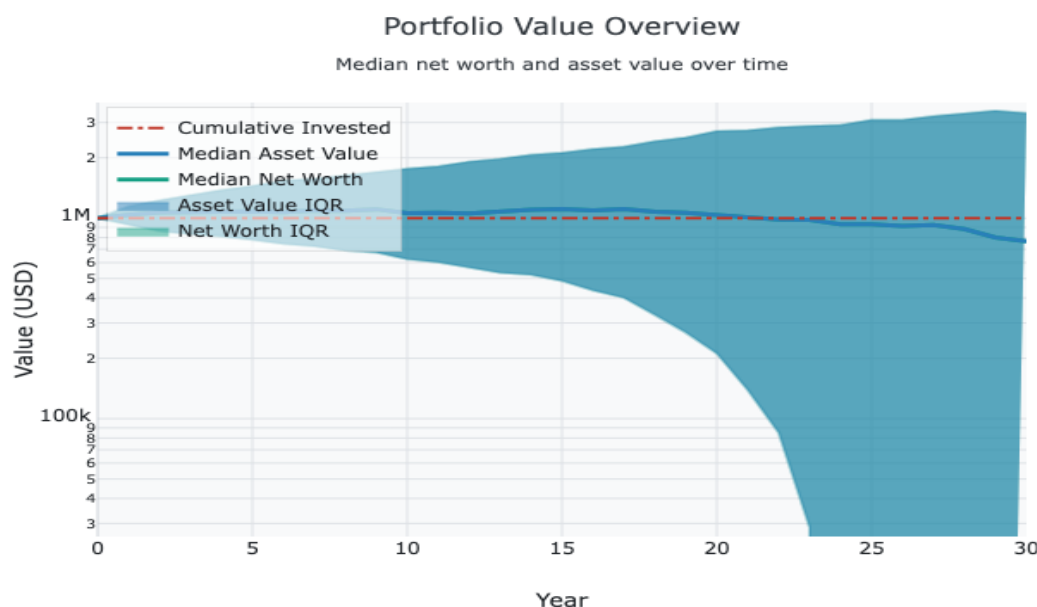


This chart visualizes the historical drawdowns of the asset, which are the percentage declines from a peak to a subsequent trough. This is a crucial measure of risk, as it quantifies the magnitude and duration of the worst-performing periods an investor would have had to endure. Understanding the historical drawdown profile is vital for setting risk tolerance and for appreciating the potential for capital loss inherent in the asset. It provides a stark reminder of the downside risk that is not always apparent from price charts alone.

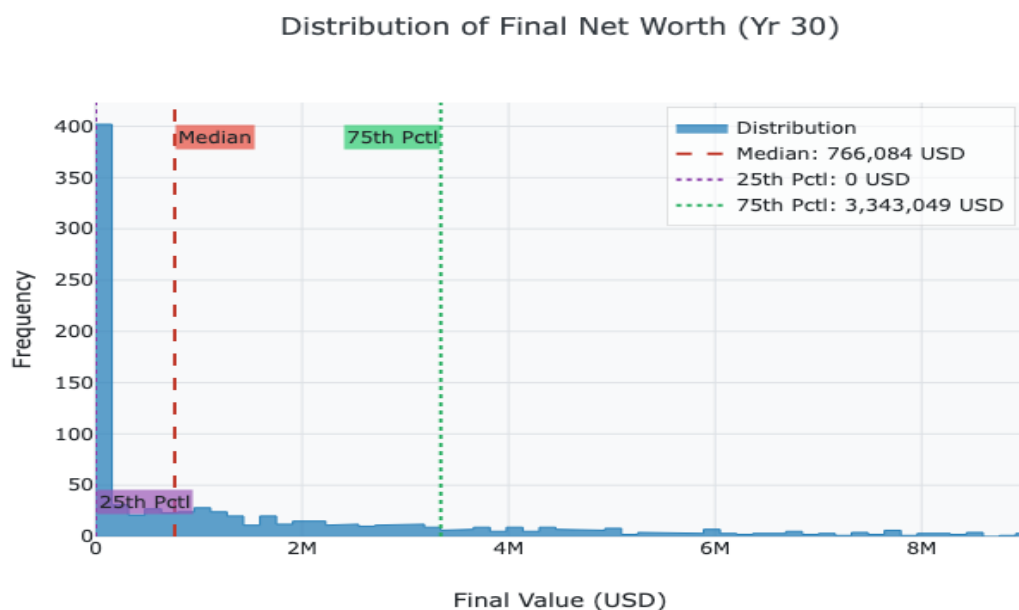
7. Simulation Summary

This chapter presents the core findings of the Monte Carlo simulation. It begins with a summary of the key outcome and risk metrics, followed by a series of visualizations that illustrate the range of potential futures. These plots show the evolution of the portfolio over time, the distribution of final outcomes, and other critical aspects of the strategy's performance.

Outcome Analysis (Yr 30)	
Median Final Net Worth	\$766,084
Median Real Net Worth (Today's USD)	\$422,932
25th Percentile Final Net Worth	\$0
75th Percentile Final Net Worth	\$3,343,049

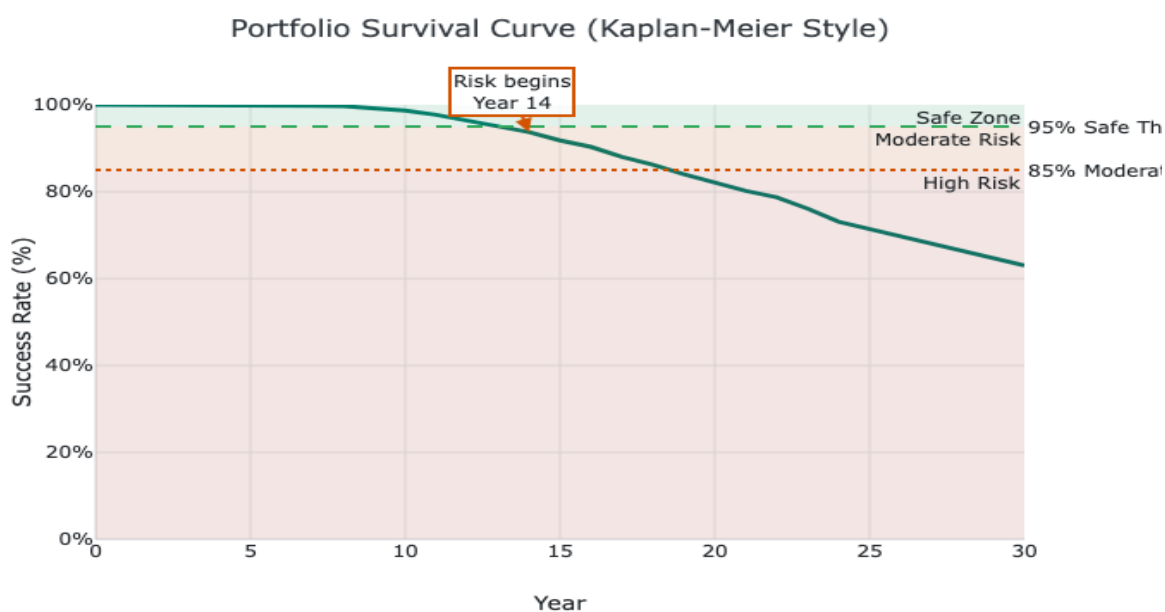


This plot shows the evolution of portfolio values over time on a logarithmic scale. It displays the median asset value and net worth as solid lines, with shaded areas representing the Interquartile Range (25th to 75th percentile) for each metric. This visualization illustrates the 'cone of uncertainty' as it widens over the years, showing how the range of possible outcomes expands with time. The logarithmic scale makes it easier to see percentage-based growth across large magnitudes.



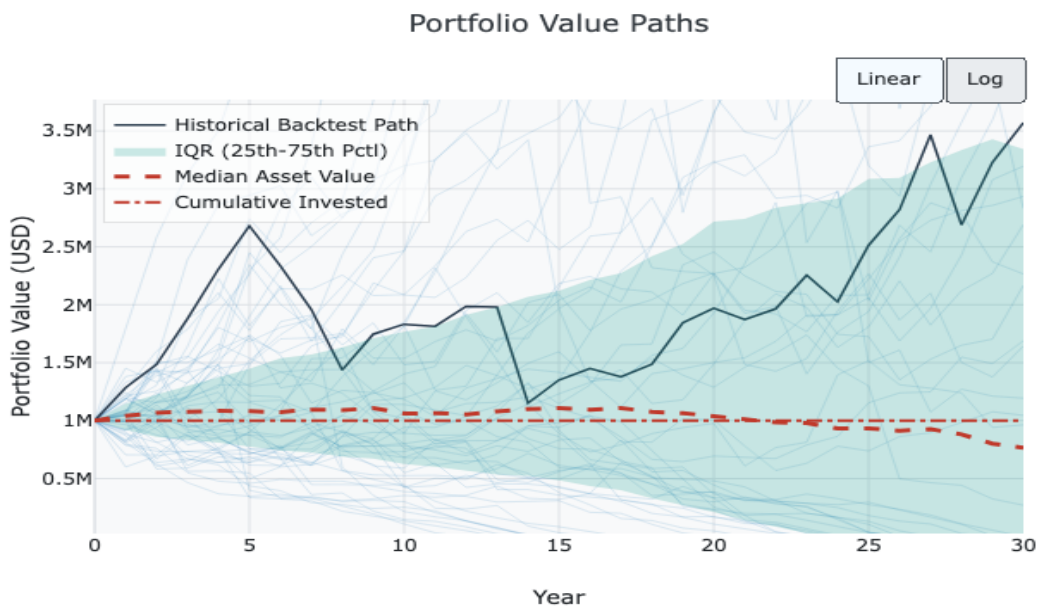
This histogram shows the distribution of final net worth (Assets - Debt) across all simulations. The x-axis is on a logarithmic scale to better represent the wide range of outcomes. This plot is crucial for understanding the probability of different results and quantifying the central tendency and spread of the final outcomes.

Risk Analysis	
Chance of Real Profit	36.3%
Chance of Ruin (Portfolio Depletion)	37.0%



This Kaplan-Meier style survival curve shows the percentage of simulations that remain successful (portfolio not depleted) at each year of the time horizon. The curve helps identify when the 'danger zone' occurs: does the risk of failure primarily happen in Year 10, Year 25, or is it spread throughout? The plot includes risk threshold lines at 95% (Safe - FIRE community standard) and 85% (Moderate Risk). Shaded zones indicate safe, moderate risk, and high risk regions. When the curve drops below 95%, it signals that the strategy has entered a zone where failure rates exceed conservative retirement planning standards. An annotation marks the first year when the success rate falls below 95%, identifying the beginning of elevated risk. A flat curve near 100% indicates a robust strategy with low failure risk throughout, while a steadily declining curve suggests accumulating risk over time, particularly from sequence-of-returns risk.

Psychological Stress Indicators	
Time Underwater	24.0 years
Recovery Time	5.0 years
Consecutive Losses (P90)	9.0 years
Severe Drawdowns	2.0 events
Years Below Initial	13.0 years

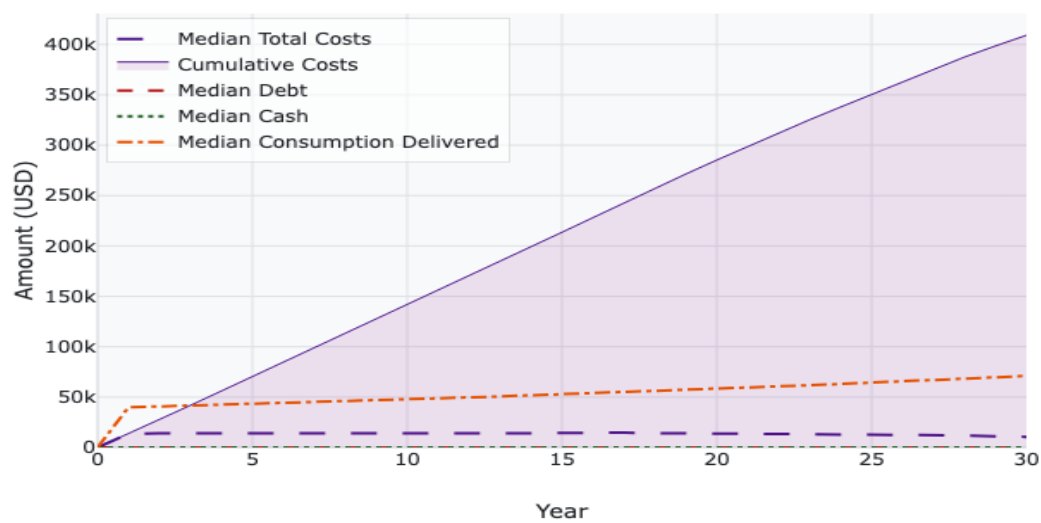


This plot displays a random subset of simulation paths for the asset value over the entire time horizon on a logarithmic scale. The faint blue lines represent individual possible futures sampled from the thousands of simulations. The shaded area shows the Interquartile Range (25th to 75th percentile). The dashed cyan line indicates the median path across all simulations. If available, the solid white line shows the historical backtest path (what actually happened using real historical data). The logarithmic scale makes it easier to see percentage-based growth, as constant growth rates appear as straight lines.

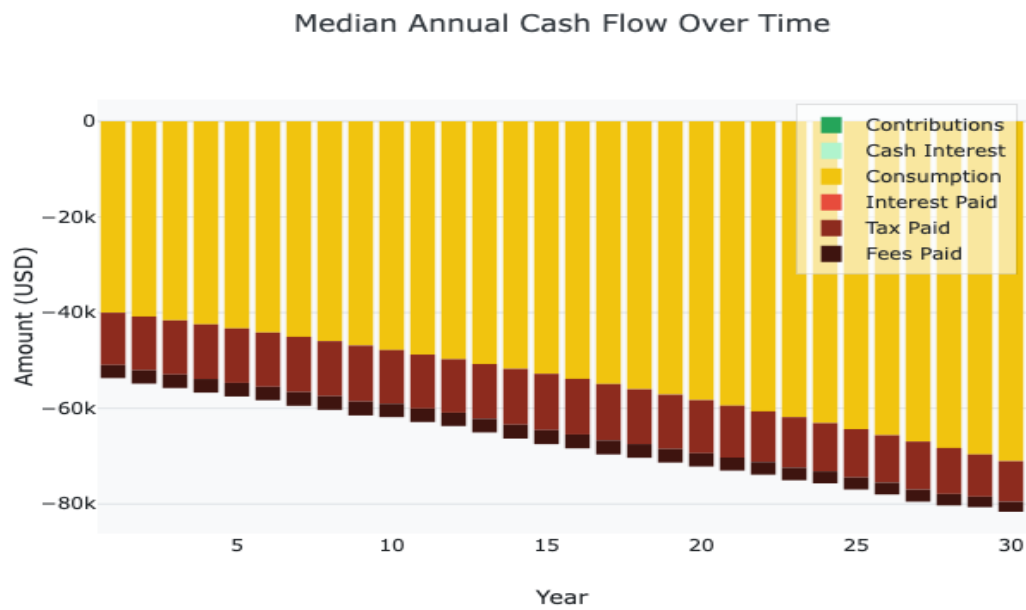
Advanced Statistics Summary (see appendix for details)

Asset Sharpe Ratio	0.41
Asset Sortino Ratio	0.58
Strategy Sharpe Ratio	-0.67
Strategy Sortino Ratio	-0.56

Cash Flow & Liabilities



This plot tracks the year-by-year cash flow mechanics and liabilities of the strategy on a linear scale. It identifies median debt levels, cash holdings, annual consumption drawdowns, total costs (taxes, fees, interest), and any contributions. By separating these details from the large-scale portfolio growth, this view provides a clear look at the engine under the hood: how liquidity is managed, how costs impact the bottom line, and how debt evolves relative to assets.



This bar chart provides a comprehensive overview of the median annual cash flows in and out of the portfolio. Positive bars (Contributions) represent new capital being added. Negative bars show capital leaving the portfolio, broken down into Consumption (money withdrawn to live on, shown in yellow) and Total Costs (the sum of all taxes, fees, and interest paid, shown in red). This plot is essential for understanding the financial mechanics of the strategy year by year, illustrating how consumption is funded and the impact of ongoing costs.

8. Qualitative Analysis

This chapter presents the detailed, narrative analysis generated by the Gemini AI model. It goes beyond the raw numbers to interpret the 'why' behind the results, discussing the interplay of different variables, potential risks, and the overall robustness of the financial strategy. This qualitative assessment provides context and depth to the quantitative findings presented elsewhere in the report.

Appendix 1: Advanced Statistics

This appendix provides definitions and the calculated values for the advanced financial metrics used in this report. These metrics offer standardized ways to measure risk and risk-adjusted return, allowing for more nuanced comparisons between different strategies and scenarios.

Asset Sharpe Ratio: 0.41

Measures the risk-adjusted return of the ****underlying asset itself****, based on its historical performance (CAGR and volatility). It ignores the strategy's cash flows and simulation outcomes.

In This Simulation: This ratio is a stable benchmark of the asset's historical risk-adjusted return. It should be comparable to the 'Sharpe Ratio' in the 'Asset Information' table, though minor differences can occur due to calculation nuances (e.g., arithmetic vs. geometric mean returns).

Real-World Context: For context, the long-term historical Sharpe Ratio for a broad market index like the S&P 500 is typically in the range of 0.4 to 0.5. This metric allows for a direct comparison to such real-world benchmarks.

Asset Sortino Ratio: 0.58

Similar to the Asset Sharpe Ratio, but only penalizes for downside volatility. It is also calculated using the asset's ****historical performance****.

In This Simulation: This provides a downside-risk-adjusted view of the underlying asset's historical performance, serving as a stable benchmark.

Real-World Context: The Sortino Ratio is almost always higher than the Sharpe Ratio. For the S&P 500, it has historically been in the 0.6 to 0.8 range.

Strategy Sharpe Ratio: -0.67

Measures the risk-adjusted return of the ****entire financial strategy****, accounting for all cash flows (e.g., withdrawals, loans). It is calculated using the Internal Rate of Return (IRR) of each simulation path.

In This Simulation: This ratio tells you how much return the overall strategy generated (above inflation) for every unit of risk taken. It is the best metric for comparing the holistic performance of one strategy versus another.

Real-World Context: This is a custom, strategy-specific metric. It is not directly comparable to the historical Sharpe Ratio of an asset, but values above 1.0 are generally considered very good.

Strategy Sortino Ratio: -0.56

An improvement on the Strategy Sharpe Ratio, this measures the strategy's excess return per unit of 'downside' risk. It only penalizes for outcomes where the annualized return was less than the risk-free rate.

In This Simulation: This answers: 'How much return did my strategy generate for only the downside risk I took on?' It is often more intuitive as it doesn't penalize for 'good' volatility (i.e.,

better-than-expected returns).

Real-World Context: Like the Strategy Sharpe, this is a custom metric. A value of 99.0 indicates no downside returns were observed in the simulation, a theoretically perfect outcome.

Value at Risk (95%): 1,000,000 SEK

A statistical measure of the potential for loss. The 95% VaR represents the maximum loss from the initial investment that is not expected to be exceeded in 95% of the simulation outcomes.

In This Simulation: This value represents the loss from your initial investment of 1,000,000 SEK that you would not expect to exceed in 19 out of 20 scenarios. It is the 5th percentile worst-case outcome in terms of capital loss.

Real-World Context: This is a common risk management metric used by banks and hedge funds to quantify potential losses. A lower VaR relative to the initial investment indicates lower tail risk.

Conditional Value at Risk (95%): 1,000,000 SEK

Also known as Expected Shortfall, CVaR goes a step further than VaR. It answers the question: 'If the 5% worst-case scenarios happen, what is the average loss?' It provides a clearer picture of the expected magnitude of losses in the tail end of the distribution.

In This Simulation: This value tells you that if you are unlucky enough to land in the worst 5% of outcomes, this is the average loss you could expect to see from your initial investment. It is always greater than or equal to the VaR.

Real-World Context: CVaR is considered a more robust measure of tail risk than VaR because it is not blind to the severity of losses beyond the VaR threshold. It is a critical metric for understanding the potential for catastrophic failure.

Appendix 2: Average Yearly Results

All values in thousands of USD unless otherwise noted.

Year	Asset Value	Cash	Debt	Net Worth	LTV	Consumption Delivered	Amount Sold	Amount C
0	1 000	0	0	1 000	0.0%	0	0	
1	1 026	0	0	1 026	0.0%	40	53	
2	1 062	0	0	1 062	0.0%	40	54	
3	1 091	0	0	1 091	0.0%	41	55	
4	1 133	0	0	1 133	0.0%	42	57	
5	1 166	0	0	1 166	0.0%	43	58	
6	1 187	0	0	1 187	0.0%	44	59	
7	1 218	0	0	1 218	0.0%	44	60	
8	1 256	0	0	1 256	0.0%	45	62	
9	1 299	0	0	1 299	0.0%	46	63	
10	1 334	0	0	1 334	0.0%	47	64	
11	1 371	0	0	1 371	0.0%	47	65	
12	1 419	0	0	1 419	0.0%	48	66	
13	1 469	0	0	1 469	0.0%	48	67	
14	1 527	0	0	1 527	0.0%	48	68	
15	1 581	0	0	1 581	0.0%	48	69	
16	1 629	0	0	1 629	0.0%	48	70	
17	1 694	0	0	1 694	0.0%	48	71	
18	1 730	0	0	1 730	0.0%	48	71	
19	1 830	0	0	1 830	0.0%	48	72	
20	1 918	0	0	1 918	0.0%	48	73	
21	1 990	0	0	1 990	0.0%	48	74	
22	2 106	0	0	2 106	0.0%	48	75	
23	2 230	0	0	2 230	0.0%	47	76	
24	2 328	0	0	2 328	0.0%	47	77	
25	2 440	0	0	2 440	0.0%	46	77	
26	2 545	0	0	2 545	0.0%	46	79	
27	2 718	0	0	2 718	0.0%	46	81	
28	2 854	0	0	2 854	0.0%	45	82	
29	2 950	0	0	2 950	0.0%	45	83	
30	3 052	0	0	3 052	0.0%	45	84	

Appendix 3: Median Yearly Results

All values in thousands of USD unless otherwise noted.

Year	Asset Value	Cash	Debt	Net Worth	LTV	Consumption Delivered	Amount Sold	Amount C
0	1 000	0	0	1 000	0.0%	0	0	
1	1 041	0	0	1 041	0.0%	40	53	
2	1 069	0	0	1 069	0.0%	40	54	
3	1 075	0	0	1 075	0.0%	41	55	
4	1 085	0	0	1 085	0.0%	42	56	
5	1 081	0	0	1 081	0.0%	43	57	
6	1 074	0	0	1 074	0.0%	44	58	
7	1 095	0	0	1 095	0.0%	45	59	
8	1 089	0	0	1 089	0.0%	45	60	
9	1 108	0	0	1 108	0.0%	46	61	
10	1 061	0	0	1 061	0.0%	47	61	
11	1 066	0	0	1 066	0.0%	48	62	
12	1 055	0	0	1 055	0.0%	49	63	
13	1 080	0	0	1 080	0.0%	50	65	
14	1 101	0	0	1 101	0.0%	51	66	
15	1 110	0	0	1 110	0.0%	52	67	
16	1 101	0	0	1 101	0.0%	53	68	
17	1 108	0	0	1 108	0.0%	54	69	
18	1 082	0	0	1 082	0.0%	56	70	
19	1 068	0	0	1 068	0.0%	57	71	
20	1 039	0	0	1 039	0.0%	58	72	
21	1 013	0	0	1 013	0.0%	59	73	
22	991	0	0	991	0.0%	60	73	
23	980	0	0	980	0.0%	61	75	
24	934	0	0	934	0.0%	63	75	
25	933	0	0	933	0.0%	64	76	
26	917	0	0	917	0.0%	65	78	
27	930	0	0	930	0.0%	66	79	
28	881	0	0	881	0.0%	68	80	
29	803	0	0	803	0.0%	69	80	
30	771	0	0	771	0.0%	71	81	

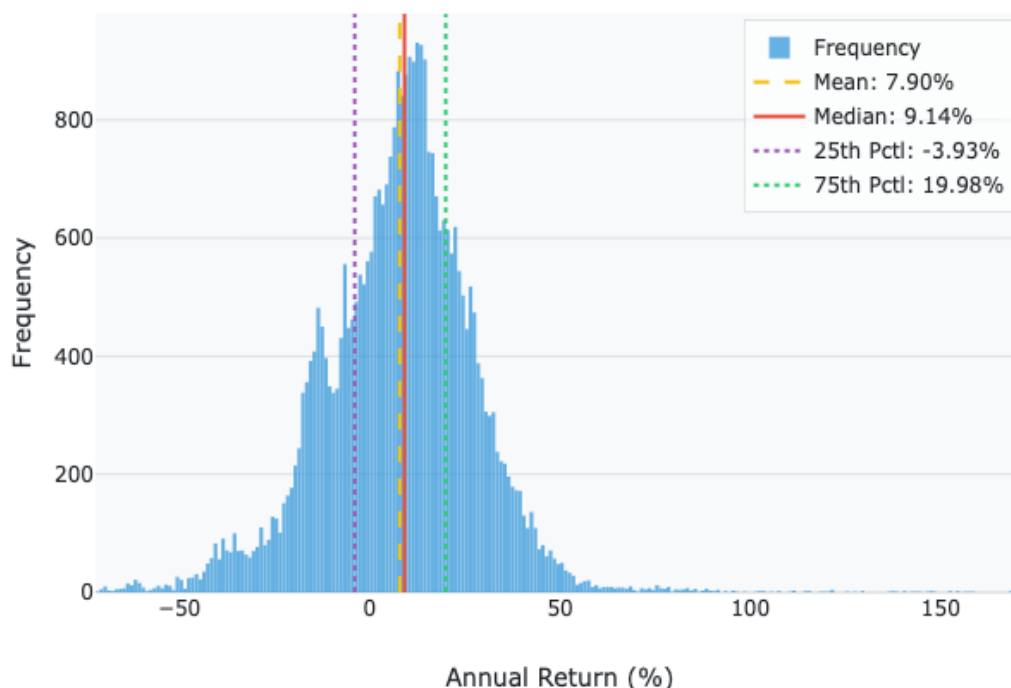
Appendix 4: Example Simulation Path

All values in thousands of USD unless otherwise noted.

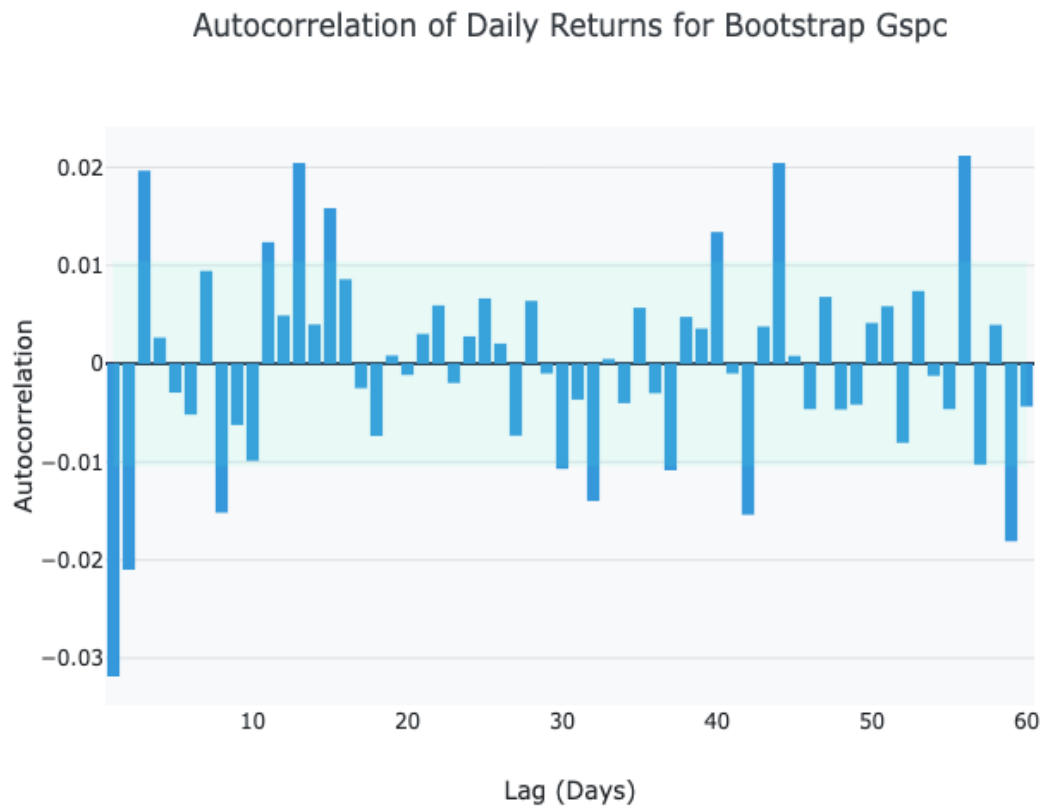
Year	Asset Value	Cash	Net Worth	Consumption Delivered	Amount Sold	Tax Paid	Fees Paid	
0	1 000	0	1 000	0	0	0	0	I
1	988	0	988	40	53	10	2	
2	994	0	994	40	53	10	2	
3	1 332	0	1 332	41	59	13	3	
4	1 530	0	1 530	42	62	15	3	
5	1 711	0	1 711	43	65	17	4	
6	1 582	0	1 582	44	64	16	4	
7	1 695	0	1 695	45	67	17	4	
8	1 722	0	1 722	45	68	17	4	
9	2 188	0	2 188	46	75	22	5	
10	2 084	0	2 084	47	74	21	5	
11	2 063	0	2 063	48	75	21	5	
12	2 235	0	2 235	49	78	23	5	
13	2 557	0	2 557	50	83	26	6	
14	2 728	0	2 728	51	86	28	7	
15	2 477	0	2 477	52	84	25	6	
16	2 670	0	2 670	53	88	27	6	
17	3 589	0	3 589	54	101	36	9	
18	4 864	0	4 864	56	118	49	12	
19	5 900	0	5 900	57	132	60	15	
20	7 148	0	7 148	58	149	72	18	
21	8 165	0	8 165	59	163	83	20	
22	9 748	0	9 748	60	184	99	24	
23	10 106	0	10 106	61	190	102	25	
24	12 375	0	12 375	63	220	125	31	
25	11 551	0	11 551	64	211	117	29	
26	9 894	0	9 894	65	191	100	25	
27	12 306	0	12 306	66	223	125	31	
28	12 212	0	12 212	68	223	124	31	
29	13 032	0	13 032	69	235	132	33	
30	15 910	0	15 910	71	273	161	40	

Appendix 5: Input Data Analysis

Distribution of Rolling 365-Day Returns (Bootstrap Gspc Data since 1927-12-30)



This histogram illustrates the distribution of rolling one-year (365-day) returns. Unlike the calendar-year returns, this method provides a much larger and more granular dataset by calculating the return for every overlapping one-year period. The resulting distribution is fundamental to the 'Bootstrap' simulation methodology. It reveals the central tendency of the asset's returns (mean/median), its dispersion (volatility), and the shape of the distribution, including its skewness and the presence of 'fat tails' (i.e., a higher likelihood of extreme outcomes than a normal distribution would suggest). This gives a probabilistic view of the returns an investor might have experienced over any given one-year period.



The autocorrelation plot (ACF) for daily returns is a critical diagnostic tool. It measures the correlation of the time series with a lagged version of itself. In financial markets, an efficient market should exhibit near-zero autocorrelation in returns, meaning past returns do not predict future returns. This plot helps to validate the assumption of a 'random walk.' Significant spikes outside the shaded confidence interval at various lags could suggest the presence of momentum or mean-reversion tendencies, which would have implications for the validity of the simulation models.

Appendix 6: Strategy Evaluations

Strategies on the leaderboard are stress-tested across **8 standardized market scenarios** to evaluate robustness and performance. This appendix explains the evaluation methodology.

Evaluation Process

Strategies on the leaderboard are stress-tested across **8 standardized market scenarios** to evaluate robustness and performance.

Simulation Settings:

- **Duration:** 30 Years (Withdrawal/Contribution) / 40 Years (Hybrid)
- **Capital:** \$1,000,000 (Withdrawal), \$100,000 (Contribution)
- **Hybrid Balancing:** Initial capital is dynamically probed per scenario to find a balanced starting point
- Target: Strategy switches from accumulation to decumulation at 50% of the simulation duration
- Tolerance: $\pm 20\%$
- **Environment:** 2% Inflation, 1% Wealth Tax (ISK)
- **Execution:** 1000 Monte Carlo simulations per scenario
- **Configuration:** Strategies are evaluated using their default parameters

Market Scenarios

1. **Bull Market** (12% return, 15% volatility) - 15% weight
2. **Normal Market** (8% return, 18% volatility) - 25% weight
3. **Bear Market** (-2% return, 20% volatility) - 20% weight
4. **High Volatility** (8% return, 30% volatility) - 15% weight
5. **Low Volatility** (5% return, 10% volatility) - 3% weight
6. **Extreme Bull** (20% return, 25% volatility) - 5% weight
7. **Extreme Bear** (-10% return, 35% volatility) - 10% weight
8. **Sideways Market** (3% return, 22% volatility) - 4% weight
9. **Stagflation** (1% return, 20% volatility) - 2% weight
10. **Deflation** (0% return, 15% volatility) - 1% weight

Note: Stress scenarios (Bear, Extreme Bear) account for 30% of the total weight.

Score Components

Excellence Score (0-100) = Weighted combination of the following metrics:

- **Present Value Score:** Lifetime Real Withdrawal Value
- **Concept:** Measures the total inflation-adjusted purchasing power extracted from the portfolio.
- **Formula:** $PV = \text{Sum of } [\text{Annual Withdrawal} / (1 + \text{Inflation})^{\text{Year}}]$
- **Normalization:** Linear scale to \$1.5M target
- **Scoring:**

Total Real Withdrawals	Score	Meaning
\$0.5M	33	Below expectations

| \$1.0M | 66 | Breakeven with initial capital |
| \$1.5M+ | 100 | Excellent - 1.5x initial |

- **Purchasing Power Score:** Inflation Tracking Capability
- **Concept:** Measures if income keeps up with cost of living (real growth of withdrawals).
- **Formula:** `Avg Year-over-Year Real Withdrawal Growth`
- **Normalization:** Linear scale with 2% real growth = 100
- **Scoring:**

Avg Real Growth	Score	Meaning
-5%	5	Severe purchasing power erosion
0%	80	Maintained purchasing power
+2%+	100	Growing lifestyle spending

- **Withdrawal Stability:** Income Predictability
- **Concept:** Measures volatility of withdrawals. Users prefer steady income over feast-and-famine.
- **Formula:** `Score = 100 - (Coefficient of Variation × 143)`
- **Normalization:** CV-based penalty (lower CV = higher score)
- **Scoring:**

CV (StdDev/Mean)	Score	Meaning
0.0	100	Perfectly stable income
0.1	86	Minor fluctuations (±10%)
0.3	57	Moderate variation (tolerable)
0.5	28	Significant swings
0.7+	0	Wild swings, hard to plan

- **Risk Score:** Portfolio Survival & Tail Risk
- **Concept:** Composite measure of not running out of money.
- **Formula:** `Score = 0.8 × Success Rate + 0.2 × Tail Risk`
- **Note:** Success Rate = % of simulations where final net worth > 0 at maturity (CVaR 95% for tail risk)
- **Scoring:**

Success Rate	Tail Risk	Score	Meaning
100%	Low	100	Never runs out
95%	Low	~85	Very safe
80%	High	~65	Risky strategy
50%	High	~40	Coin flip

- **Robustness Score:** All-Weather Consistency
- **Concept:** Penalizes strategies that only work in specific market conditions.
- **Formula (Withdrawal/Hybrid):** `0.4 × Worst Case Sortino + 0.3 × Volatility + 0.3 × Consistency`
- **Formula (Accumulation):** `0.4 × Worst Drawdown + 0.3 × Drawdown Consistency + 0.3 × Ulcer

Index`

- **Scoring:**

Worst/Best Ratio	Consistency	Score	Meaning
----- ----- ----- -----			
0.9+	High	90+	Works everywhere
0.5	Medium	~60	Some variability
0.2	Low	~30	Bull market only

- **Capital Efficiency:** Lifetime Consumption Efficiency

- **Concept:** Total real withdrawals vs total real inputs (excludes legacy).

- **Formula:** `Real Withdrawals / (Initial + Contributions)`

- **Key difference:** Unlike Legacy Score (terminal wealth), this measures how much you actually consumed.

- **Normalization:** Sigmoid centered at 2.0x

- **Scoring:**

Efficiency Ratio	Score	Meaning
----- ----- -----		
0.5x	0	Got back less than half
1.0x	~12	Consumed what you put in
2.0x	~50	Consumed double
3.0x	~85	Consumed triple
5.0x+	100	Exceptional consumption

- **Legacy Score:** Capital Preservation

- **Concept:** Real wealth remaining at end vs total investment.

- **Formula:** `Final Real Net Worth / Total Real Invested`

- **Normalization:** Sigmoid centered at 0.75x

- **Scoring:**

Preservation Ratio	Score	Meaning
----- ----- -----		
0%	2	Spent everything
50%	~18	Half remaining
75%	~50	Most preserved
100%	~88	Fully preserved
150%+	~98	Grew principal

- **Consumption Ratio:** Pure Withdrawal Efficiency

- **Concept:** Measures how much you consumed vs what you invested (excludes legacy).

- **Formula:** `Ratio = Real\_Withdrawals / (Initial + Real\_Contributions)`

- **Key difference:** Unlike Capital Efficiency, this does NOT reward leaving money behind.

- **Normalization:** Sigmoid curve centered at 1.5x

- `Score = 100 / (1 + e^(-2.0 × (ratio - 1.5)))`

- **Scoring:**

Consumption Ratio	Score	Meaning
-------------------	-------	---------

ROI Ratio	Score	Meaning
1.0x	27	Breakeven (withdrew what you put in)
1.5x	50	Consumed 50% more than invested
2.5x	88	More than doubled consumption
4.0x+	100	Excellent market leverage

- **Coast FIRE Score:** Coast FIRE Timeline
- **Concept:** How quickly does growth exceed contributions? (Coast FIRE milestone)
- **Philosophy:** The faster you reach a point where you can stop contributing and still hit retirement goals, the more financially free you are.
- **Formula:** $\text{ROI Ratio} = \text{Final Real Wealth} / (\text{Initial} + \text{Real Contributions})$
- **Scoring:**

ROI Ratio	Score	Meaning
1.0x	0	Breakeven only
2.0x	25	Doubled invested capital
3.0x	50	Tripled invested capital
5.0x+	100	Exceptional growth (5x or more)

- **Accumulation Velocity:** Wealth Accumulation Rate
- **Concept:** How fast does wealth grow per year? (CAGR)
- **Formula:** $\text{CAGR} = (\text{Final} / \text{Initial})^{(1/\text{Years})} - 1$
- **Scoring:**

CAGR	Score	Meaning
0%	0	No real growth
5%	50	Moderate growth (inflation + 3%)
10%+	100	Exceptional growth

- **Contribution Efficiency:** Contribution ROI
- **Concept:** How much wealth was generated per dollar contributed?
- **Formula:** $(\text{Final Wealth} - \text{Initial}) / \text{Total Real Contributions}$
- **Scoring:**

Multiplier	Score	Meaning
1.0x	25	Contributions preserved value
2.0x	50	Doubled contribution value
4.0x+	100	Quadrupled or more

- **Sharpe Ratio:** Risk-Adjusted Return (Sharpe Ratio)
- **Concept:** Excess return per unit of total volatility. Industry standard for comparing performance.
- **Formula:** $(\text{Strategy Return} - \text{Risk-Free Rate}) / \text{Volatility}$
- **Key difference:** Unlike Risk Score (survival probability), Sharpe measures efficiency of risk-taking.

- **Scoring:**

Sharpe Ratio	Score	Meaning
< 0	0	Losing money on risk-adjusted basis
0.5	25	Below average
1.0	50	Good risk-adjusted return
2.0+	100	Excellent (hedge fund territory)

- **Calmar Ratio:** Return vs Drawdown (Calmar Ratio)

- **Concept:** Annualized return divided by maximum drawdown. Higher = better risk-return tradeoff.

- **Formula:** $\text{Annualized Return} / |\text{Max Drawdown}|$

- **Key difference:** Unlike Robustness (consistency), Calmar measures return per unit of worst-case pain.

- **Scoring:**

Calmar Ratio	Score	Meaning
< 0	0	Negative returns
1.0	33	Moderate risk-return
2.0	67	Good risk-return
3.0+	100	Excellent (minimal drawdown for return)

- **Downside Stability:** Downside Income Stability

- **Concept:** Measures only DOWNSIDE withdrawal volatility (ignores positive deviations).

- **Formula:** $\text{Score} = 100 - (\text{Downside CV} \times 200)$

- **Key difference:** Unlike Withdrawal Stability, does NOT punish positive income surprises.

- **Scoring:**

Downside CV	Score	Meaning
0.0	100	No downside income variation
0.1	80	Minor income dips
0.25	50	Moderate downside risk
0.5+	0	Severe income instability

- **Ulcer Index:** Drawdown Pain (Ulcer Index)

- **Concept:** Measures depth AND duration of drawdowns. Higher Ulcer Index = more pain.

- **Formula:** $\text{Score} = 100 - \text{Ulcer Index}$

- **Key difference:** Unlike Stability (income smoothness), Ulcer measures portfolio-level drawdown suffering.

- **Scoring:**

Ulcer Index	Score	Meaning
0-10	90-100	Minimal drawdown pain
20-30	70-80	Moderate drawdowns

| 50 | 50 | Significant drawdown experience |
| 100+ | 0 | Extreme and prolonged drawdowns |

■ Wisdom of the Crowd

The leaderboard is a **collective intelligence engine**. Research shows that diverse groups often outperform individual experts when:

- There's a **clear scoring mechanism** (Excellence Score)
- Contributors are **independent** (users from different backgrounds)
- Results are **aggregated effectively** (the leaderboard)

As thousands of users create and refine strategies, the top performers represent the **collective wisdom** of the community—potentially discovering approaches that no single financial advisor would design alone.

■ *Based on ["The Wisdom of Crowds" (Surowiecki, 2004)](https://en.wikipedia.org/wiki/The_Wisdom_of_Crowds)*

Appendix 7: Glossary

Core Simulation & Statistical Concepts

Annualized Volatility

The standard deviation of returns scaled to represent a full year. It measures the degree of variation in an asset's returns over a one-year period, providing a standardized measure of risk that can be compared across different assets and time periods.

Autocorrelation (ACF)

The correlation of a time series with a delayed version of itself. In finance, the ACF of returns is used to test for market efficiency. If returns are random (an efficient market), the autocorrelation for any lag other than zero should be close to zero. Significant spikes outside the shaded confidence interval at various lags could suggest the presence of momentum or mean-reversion tendencies, which would have implications for the validity of the simulation models.

Cone of Uncertainty

A visual representation in simulation plots showing how the range of possible outcomes widens over time. It illustrates the increasing uncertainty about future values as the time horizon extends, typically shown as shaded areas between percentile paths.

Fat Tails

A characteristic of probability distributions where extreme events (far from the mean) occur more frequently than predicted by a normal distribution. In financial markets, fat tails mean that crashes and booms are more common than standard models suggest, making risk management more challenging.

Kurtosis

A measure of the 'tailedness' of a probability distribution. High kurtosis indicates heavy tails and a higher probability of extreme outcomes (both positive and negative) compared to a normal distribution. This is crucial for understanding the risk of rare but severe market events.

Logarithmic Scale

A non-linear scale used for displaying data that spans several orders of magnitude. On a log scale, constant percentage changes appear as straight lines, making it easier to visualize compound growth and relative changes.

Monte Carlo Simulation

A method used to understand the impact of risk and uncertainty in financial forecasts. It works by running a large number of simulations with random variables to see the range of possible outcomes.

Sequence of Returns Risk

The risk that the order and timing of investment returns are unfavorable. Poor returns in the early years of retirement can have a much bigger impact than poor returns later on.

Skewness

A measure of the asymmetry of a probability distribution. Positive skewness indicates a distribution with a longer tail on the right side (more extreme positive outcomes), while negative skewness indicates a longer tail on the left (more extreme negative outcomes). In finance, understanding skewness helps assess the likelihood of extreme gains or losses.

Standard Deviation

A statistical measure of the amount of variation or dispersion of a set of values. In finance, it's often used as a measure of volatility.

Volatility

A measure of how much the price of an asset varies over time. Higher volatility means higher risk and a wider range of potential outcomes. It is a key driver of both upside potential and downside risk.

Financial Strategies & Methods

4% Rule

A rule of thumb suggesting that retirees can safely withdraw 4% of their initial portfolio value each year, adjusted for inflation, without depleting their funds for at least 30 years.

Asset Withdrawal

A strategy for generating cash by selling assets from a portfolio. This is a common strategy for retirees.

Backtest

A simulation run using actual historical data to see how a strategy would have performed in the past. Unlike forward-looking Monte Carlo simulations, a backtest uses the exact sequence of historical returns to provide a single 'what actually happened' scenario.

Loan Drawdown (Buy, Borrow, Die)

A strategy where instead of selling assets, one borrows money against the value of the portfolio to fund expenses. The assets remain invested, allowing them to continue to grow over time. The 'die' part refers to the idea that the loan is repaid from the estate upon death, thus avoiding capital gains tax on the assets.

Trinity Study

A well-known financial study that examined 'safe withdrawal rates' from retirement portfolios. It's famous for establishing the '4% Rule'.

Portfolio Outcome & Risk Metrics

Cash Flow Reliability

A qualitative measure of the consistency of annual cash flow. A 'Low' reliability indicates that consumption drawdowns were frequently suspended (e.g., due to high LTV) or reduced (e.g., due to a percentage-based drawdown on a declining asset value).

Chance of Profit (Nominal)

The percentage of simulations where the final net worth is greater than the initial investment in nominal terms (not adjusted for inflation).

Chance of Real Profit

The percentage of simulations where the final net worth is greater than the initial investment after adjusting the initial amount for cumulative inflation over the simulation period. This measures the growth of purchasing power.

Chance of Ruin

The inverse of Portfolio Success Rate - the percentage of simulations that failed to sustain the strategy for the full time horizon. For withdrawal strategies, this means portfolio depletion (running out of money). For leverage strategies, this means insolvency (debt exceeding asset value). Also called the failure rate.

Earliest Ruin Year

Among the simulations that failed, this indicates the year when the first failure occurred. This represents the fastest path to ruin and helps understand the minimum time buffer before failure becomes possible, even in worst-case sequences of returns.

Final Net Worth (Real | Nominal)

The portfolio's net worth at the end of the simulation, shown in both real (inflation-adjusted to today's purchasing power) and nominal (actual future dollars) terms. This dual display helps you understand both the actual amount you'd have and what it would be worth in today's money.

Maximum Drawdown

The largest peak-to-trough decline in asset value during the historical period or simulation. It represents the worst-case scenario an investor would have experienced and is a key measure of downside risk. Expressed as a percentage of the peak value.

Median Year of Ruin

For the set of simulations that end in failure (portfolio depletion or insolvency), this is the median year in which that failure occurred. It provides insight into how quickly things can go wrong in unfavorable scenarios.

Net Worth

The value of assets minus liabilities (debts). In this simulation, it's the asset value minus the total debt. This represents the owner's equity in the portfolio.

Peak Leverage (95th %ile)

The highest Loan-to-Value (LTV) ratio reached across 95% of simulation paths. This indicates how much borrowing pressure the strategy experiences in typical scenarios. The 95th percentile excludes the most extreme 5% of outcomes to focus on reasonably likely scenarios rather than tail events.

Portfolio Success Rate

The percentage of simulations that successfully funded consumption for the entire time horizon without failure. For the Trinity strategy, failure is defined as portfolio depletion (value reaching zero). For the Loan Drawdown strategy, failure is defined as insolvency (debt exceeding asset value).

Return on Investment

The total percentage gain or loss on the initial investment, calculated as $(\text{Final Value} - \text{Initial Investment}) / \text{Initial Investment}$. Can be shown as total return or annualized (CAGR) for comparing across different time periods.

Risk of Insolvency

The risk that liabilities (debt) will exceed the value of assets, resulting in a negative net worth.

Risk of Nominal Loss

The percentage of simulations where the final net worth is less than the initial investment, without adjusting for inflation. It measures the risk of ending with less money than you started with in absolute terms.

Total Costs

The sum of all fees, taxes, and interest paid over the simulation period. This includes asset management fees, capital gains taxes, wealth taxes, and loan interest costs. Represents the total drag on portfolio performance from operating expenses.

Total Portfolio Value (Real | Nominal)

For withdrawal strategies, this is the sum of final net worth plus all withdrawals taken, shown in both inflation-adjusted and nominal terms. Represents the total value generated by the portfolio over its lifetime.

Total Withdrawn

The cumulative sum of all consumption withdrawals taken from the portfolio over the simulation period, adjusted for inflation to today's purchasing power. This represents the total lifestyle funding that the portfolio provided.

Worst 5% Outcome

The final net worth at the 5th percentile - meaning only 5% of simulations ended worse than this. Represents a pessimistic but plausible worst-case scenario that helps with downside risk assessment.

Years of Spending Left (Portfolio Longevity)

Estimates how many additional years the portfolio could sustain withdrawals based on the final net worth and a representative withdrawal amount. Calculated as: final net worth divided by the median of all non-zero, inflation-adjusted withdrawals during the simulation. All historical withdrawals are adjusted to final-year purchasing power to ensure an accurate comparison. For withdrawal/hybrid strategies, this uses the median withdrawal across all years with actual withdrawals. For contribution-only strategies, this metric is not applicable (displayed as N/A). This approach avoids misleading results from strategies that intentionally reduce withdrawals near the end of the simulation (e.g., 'die with zero' approaches).

■■ Cost Drag

The percentage of total portfolio value lost to fees, taxes, and interest costs. Shows how much of your potential wealth was consumed by operating expenses. Particularly important for the FIRE community focused on minimizing unnecessary costs.

Transaction Types & Strategy Actions

Asset Purchases

Buying additional assets with available cash. Increases portfolio value and future growth potential. Common in accumulation phases or when rebalancing after deleveraging.

Asset Sales

Selling portfolio assets to generate cash. Used to fund withdrawals, pay costs, or reduce leverage. Triggers capital gains tax on any profit since purchase. Common in withdrawal-based strategies like the Trinity approach.

Borrowing

Taking a loan against the portfolio's value to generate cash for expenses without selling assets. Incurs interest costs but avoids capital gains tax and keeps assets invested. Core mechanic of the Buy, Borrow, Die strategy.

Cash Buffer Management

Maintaining cash reserves to fund near-term withdrawals without selling assets during market downturns. Helps avoid selling at unfavorable prices (sequence of returns risk mitigation).

Contributions

Adding new capital to the portfolio from external sources (salary, savings, etc.). Increases portfolio value without requiring market gains. Critical during accumulation phases.

Debt Repayment

Paying down existing loans to reduce interest costs and improve the loan-to-value ratio. Reduces financial leverage and risk but requires cash or asset sales to fund.

Deleveraging

Emergency asset sales triggered when loan-to-value ratios exceed safety thresholds. Reduces debt to restore healthy leverage ratios. A critical automatic risk management mechanism in leveraged strategies.

Strategy Evaluation Metrics

Capital Efficiency

Measures how efficiently the strategy uses capital to achieve its goals. Considers factors like tax efficiency, fee minimization, and optimal leverage use. Higher scores indicate smarter use of resources.

Legacy Score

Evaluates the potential to pass wealth to heirs. Higher final net worth and lower failure rates contribute to this score. Relevant for those prioritizing intergenerational wealth transfer.

Present Value Score

Measures the median final net worth relative to a target value (typically the initial investment). Higher scores indicate better wealth preservation. Weighted heavily in balanced evaluations as it directly measures the primary outcome.

Purchasing Power Score

Evaluates how well the strategy maintains real (inflation-adjusted) wealth. Accounts for the erosion of purchasing power over time. Critical for long-term financial security.

Risk Score

A composite measure of downside risks including failure rates, drawdowns, and leverage extremes. Lower risk scores are better. Heavily weighted in conservative evaluation profiles.

Robustness Score

Evaluates how well a strategy performs across diverse market scenarios (bull, bear, sideways, volatile). Higher scores indicate strategies that work in many conditions, not just favorable ones.

Withdrawal Stability

Measures the consistency and reliability of consumption withdrawals over time. Higher scores indicate more predictable cash flows. Important for retirement planning where stable income is valued.

Asset & Market Information

Asset Name

The descriptive name of the investment asset used in the simulation (e.g., 'S&P; 500', 'Bitcoin'). Identifies what historical data or model parameters were used.

End Date

The final date of the historical price data. Represents the most recent market conditions incorporated into the simulation model.

Start Date

The beginning date of the historical price data used for the simulation. Earlier start dates provide more data for bootstrap sampling but may include different market regimes.

Ticker

The market ticker symbol for the asset (e.g., 'BTC', '^GSPC'). Used to identify and fetch historical price data from financial APIs.

Total Trading Days

The number of individual trading days in the historical dataset. More trading days provide a richer dataset for statistical analysis and bootstrap sampling.

Simulation Settings & Parameters

Asset Model

The source of return data for the simulation. Either 'Bootstrap' (randomly samples from historical returns) or 'Parametric' (generates returns from a statistical model with specified mean and volatility).

Capital Gains Tax Rate

The tax rate applied to profits when assets are sold. Only charged on the gain (sale price minus cost basis), making it more efficient than wealth taxes for long-term holders.

Cash Interest Rate

The annual interest rate earned on cash holdings. Applies to the cash buffer maintained in the portfolio. Typically set to a risk-free rate like short-term Treasury yields.

Inflation Rate

The assumed annual inflation rate used to adjust values to real (today's purchasing power) terms. Critical for understanding true wealth preservation over long time horizons.

Initial Investment

The starting capital amount invested at the beginning of the simulation (Year 0). All returns, withdrawals, and outcomes are relative to this baseline amount.

Isk Tax Rate

The annual Swedish wealth tax rate applied to the portfolio value (Investeringssparkonto system). Charged yearly regardless of whether assets are sold or gains realized.

Loan Interest Rate

The annual interest rate charged on portfolio loans (for leverage strategies). Higher rates increase the cost of borrowing and can significantly impact leveraged strategy viability.

Num Simulations

The number of Monte Carlo simulation paths to run. More simulations provide more statistical confidence but take longer to compute. Typically 1,000-10,000 for good results.

Num Years

The time horizon for the simulation in years. Represents how long you want to test the strategy (e.g., 30 years for a typical retirement period).

Return Threshold Rate

For bootstrap models with return filtering, this sets a maximum return cap. Returns above this threshold are excluded to create more conservative scenarios that don't rely on exceptional bull markets.

Simulation Name

A user-provided identifier for this simulation run. Helps organize and distinguish between different scenarios when reviewing historical results.

Strategy

The decision-making logic that determines how the portfolio behaves each year (withdrawals, contributions, borrowing, etc.). Can be a built-in strategy like Trinity or Buy-Borrow-Die, or a custom

user-designed strategy.

Tax Method

Which taxation system is used: 'ISK' (annual wealth tax) or 'Capital Gains' (tax on realized profits). Different methods significantly impact strategy performance and optimal behavior.

Info Box Metrics

25th Percentile Final Net Worth

The final net worth at the 25th percentile - meaning 75% of simulations ended better than this. Represents a moderately pessimistic outcome for downside planning.

75th Percentile Final Net Worth

The final net worth at the 75th percentile - meaning only 25% of simulations ended better. Represents a moderately optimistic outcome for upside potential assessment.

Asset Sharpe Ratio

Risk-adjusted return of the underlying asset based on historical data. Calculated as (return - risk-free rate) / volatility. Provides a baseline benchmark independent of strategy.

Asset Sortino Ratio

Like Sharpe Ratio but only penalizes downside volatility, not upside. More intuitive as investors generally don't mind upside volatility. Higher is better.

Chance of Ruin (Insolvency)

For leverage strategies, the percentage of simulations where debt exceeded asset value (negative net worth), resulting in insolvency and strategy failure.

Consecutive Losses (P90)

At the 90th percentile (near-worst case), the longest streak of consecutive years with portfolio value declines. Extended losing streaks severely test psychological resilience.

Median Final LTV

The median loan-to-value ratio at the end of the simulation across all paths. Shows typical debt levels in leveraged strategies. Lower values indicate safer final leverage states.

Median Final Net Worth

The middle value (50th percentile) of final net worth across all simulations. Half of outcomes are above this, half below. Preferred over mean as it's less affected by extreme outliers.

Median Real Net Worth (Today's SEK)

The median final net worth adjusted for inflation to express its value in today's purchasing power. Shows what the portfolio would actually be worth in current terms.

Recovery Time

The median time (in years) required to recover from the worst drawdown experienced during the simulation. Long recovery times test investor patience and discipline.

Severe Drawdowns

The median number of distinct 20%+ drawdown events experienced. Frequent severe corrections indicate high psychological volatility even if the strategy eventually recovers.

Strategy Sharpe Ratio

Risk-adjusted return of the complete strategy including all cash flows (withdrawals, contributions, costs). Shows how efficiently the strategy generated returns per unit of risk taken.

Strategy Sortino Ratio

Downside-focused risk-adjusted return for the complete strategy. Particularly relevant for strategies where avoiding losses is more important than maximizing gains.

Years Below Initial

The median number of years the portfolio value spent below the initial investment amount. Being 'underwater' triggers loss aversion and regretful decision-making.

Performance & Risk-Adjusted Return Ratios

CAGR (Compound Annual Growth Rate)

The mean annual growth rate of an investment over a specified period longer than one year. It represents the rate at which an investment would have grown if it had grown at a steady rate, smoothing out volatility. Calculated as: $(\text{Ending Value} / \text{Beginning Value})^{(1/\text{Years})} - 1$.

Calmar Ratio

A risk-adjusted return metric calculated as the CAGR divided by the maximum drawdown. It measures the return per unit of downside risk. A higher Calmar Ratio indicates that the investment generated strong returns relative to its worst historical decline, making it useful for assessing strategies with different drawdown profiles.

Sharpe Ratio

A measure of risk-adjusted return, calculated as the excess return (return above the risk-free rate) divided by the standard deviation of returns. A higher Sharpe Ratio indicates better risk-adjusted performance. It answers the question: 'How much extra return am I getting for each unit of risk I'm taking?'

Sortino Ratio

Similar to the Sharpe Ratio, but it only considers downside volatility (negative returns) rather than total volatility. This makes it a more relevant measure for investors who are primarily concerned with downside risk. A higher Sortino Ratio indicates better downside risk-adjusted performance.

LTV & Deleveraging Concepts

Deleveraging

The act of reducing debt. In this simulation, it is an automated process triggered when the LTV exceeds the 'Action Tier' threshold, forcing the sale of assets to pay down the loan to a 'Target LTV'. This is a critical risk-mitigation mechanism.

LTV Action Threshold

The second, more critical tier in the LTV management system. When LTV exceeds this threshold, the system triggers automatic deleveraging by selling assets to pay down debt. This is an emergency measure to prevent insolvency.

LTV Warning Threshold

The first tier in the LTV management system. When LTV exceeds this threshold, new loan drawdowns are suspended to prevent further leverage increase. This is a proactive risk management measure designed to prevent the LTV from reaching dangerous levels.

Loan-to-Value (LTV)

The ratio of a loan to the value of the asset purchased or used as collateral. A higher LTV means higher leverage and higher risk.

Maximum LTV

The highest Loan-to-Value (LTV) ratio reached at any point during a single simulation run. The distribution of this metric across all simulations is a key indicator of the strategy's leverage risk.

Target LTV

The LTV level that the deleveraging process aims to achieve when selling assets to reduce debt. Typically set below the Action Threshold to provide a safety buffer and prevent immediate re-triggering of deleveraging.

Tiered LTV Management

A risk management system with multiple thresholds. In this simulation, it includes a 'Warning Tier' that suspends new drawdowns and an 'Action Tier' that triggers deleveraging (asset sales) to reduce debt.

Data & Return Concepts

Bootstrap (Simulation Model)

A simulation method that uses historical data. It involves randomly sampling from historical returns to generate future return scenarios. It preserves the historical distribution of returns.

Drawdown

This term has two meanings in the report. 1) **Market Drawdown:** The peak-to-trough decline of an investment's market value, measuring downside risk. 2) **Consumption Delivered:** The amount of money taken from the portfolio for consumption, either by selling assets or by taking a loan.

Parametric (Simulation Model)

A simulation method that uses statistical parameters (like mean and standard deviation) derived from historical data or assumptions. Returns are generated from a specified probability distribution (e.g., a normal distribution).

Percentile Paths

In simulation visualizations, these are lines representing specific percentiles of outcomes across all simulations. For example, the '10th percentile path' shows how the portfolio evolved in the worst 10% of scenarios, while the '90th percentile path' shows the best 10% of scenarios.

Rolling Return

The average annualized return for a period ending at a particular date. For example, a 'rolling 1-year return' is calculated for every day of a dataset, providing a more comprehensive view of performance than calendar-year returns.

Market Efficiency & Behavior Concepts

Efficient Market

A market where asset prices fully reflect all available information, making it impossible to consistently achieve returns above the market average through stock picking or market timing. In an efficient market, returns should follow a random walk.

Mean Reversion

A tendency for asset prices or returns to move back toward their historical average over time. Assets that have experienced extreme gains may decline, and those with extreme losses may recover. Mean reversion would show up as negative autocorrelation in returns.

Momentum

A tendency for assets that have performed well in the recent past to continue performing well in the near future, and vice versa. The presence of momentum would show up as positive autocorrelation in returns and challenges the efficient market hypothesis.

Random Walk

A theory suggesting that stock price changes are random and unpredictable, with each price movement independent of previous movements. This is a key assumption in many financial models and implies that past price movements cannot be used to predict future movements.

Psychological & Behavioral Metrics**Consecutive Declining Years (P90)**

The longest streak of consecutive years where the portfolio value declined year-over-year. This metric captures sustained losing periods that severely test psychological resilience. Most investors struggle to maintain conviction after 2-3 consecutive down years, regardless of long-term fundamentals. Reported as the 90th percentile (worst-case scenario) to highlight extreme stress periods.

Peak Drawdown Recovery Time

The number of years required to recover from the worst drawdown (peak-to-trough decline) experienced during the simulation. This measures how long an investor must wait to 'break even' after their worst-case scenario. Research shows that recovery periods exceeding 3-5 years significantly increase the likelihood of investors abandoning their strategy. Reported as the median across all simulation paths.

Severe Drawdown Frequency

The number of distinct periods where the portfolio experienced a drawdown of 20% or more from a peak. This counts major market corrections and crashes—events that trigger intense emotional responses and media panic. Frequent severe drawdowns indicate high psychological volatility, even if the strategy eventually recovers. Reported as the median across all simulation paths.

Time Underwater

The number of years a portfolio spends below its previous all-time peak value. This metric quantifies how long an investor must endure seeing their portfolio 'underwater'—a psychologically challenging state that often triggers panic selling. Extended underwater periods test investor discipline and are a primary driver of strategy abandonment, even for fundamentally sound approaches. Reported as the median across all simulation paths.

Years Below Initial Investment

The number of years where the portfolio value remains below the starting capital. Being 'down' on initial investment creates powerful regret and loss aversion, often prompting investors to 'cut their losses' prematurely. This metric is particularly relevant for retirees who may not have time to wait for recovery. Reported as the median across all simulation paths.

Taxation Concepts**Asset Management Fee**

An annual fee charged by a fund manager or investment service, calculated as a percentage of the total asset value. In this simulation, it is treated as an additional expense that must be covered either by selling assets (Trinity) or by increasing debt (Loan Drawdown).

Capital Gains Tax

A tax on the profit realized on the sale of a non-inventory asset. It's the difference between the sale price and the original cost (cost basis).

Cost Basis

The original value of an asset for tax purposes, usually the purchase price. It's used to calculate capital gains. In this simulation, the cost basis of the portfolio is tracked to accurately model the tax implications of deleveraging sales.

Wealth Tax / ISK Tax

A Swedish tax system (Investeringsparkonto) where a small annual tax is applied to the total value of the investment account, rather than taxing capital gains when assets are sold. This simplifies taxation by charging a fixed percentage of the account value each year, regardless of whether gains or losses were realized. It's an alternative to the traditional capital gains tax model.

General Statistical Terms

Inflation-Adjusted Value (Real Value)

The value of money or assets expressed in terms of the amount of goods or services that one unit of money can buy. It's the value after accounting for inflation.

Interquartile Range (IQR)

A measure of statistical dispersion, being equal to the difference between the 75th and 25th percentiles. It represents the range of the middle 50% of outcomes and is less sensitive to extreme outliers than the standard deviation.

Median

The middle value in a sorted list of numbers. 50% of the outcomes are above the median and 50% are below.

Nominal Value

The face value of money or assets, without being adjusted for inflation.

Percentile

A measure used in statistics indicating the value below which a given percentage of observations in a group of observations falls. For example, the 10th percentile is the value below which 10% of the observations may be found.

Appendix 8: AI Prompt